

Zoravur Singh

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EXPERIENCE

Solutions That Matter

Software Engineer

Toronto, ON

Jan 2024 - Present

- Consolidated order entry methods across the company by delivering a custom order processing system end-to-end.
- Saved an estimated ~120 hours/week by reducing processing errors, customer escalations, and status inquiries.

Royal Bank of Canada

Toronto, ON

AI Engineering Intern

Sep 2021 - Mar 2022

- Contributed to Aiden, a RL trading algorithm handling more than \$100 million daily by adding new trading signals.
- Built a C++ market exchange simulator, reducing training cycles from 3 days to 14 hours.

PROJECTS

Manipulation Playground Apart Research hackathon paper on emergent LLM-to-LLM manipulation.

Jan 2026

 github.com/zoravur/manipulation-game

- Co-authored an Apart Research paper on emergent LLM-to-LLM manipulation, extending Cheap Talk (Pham 2025).
- Ran a 10-pair matchup of five frontier models, with an LLM judge scoring scheming and manipulation success.
- Reported Gemini 3 Flash manipulating unprompted; Claude Sonnet 4.5 manipulated GPT-5.2 less than GPT-5-mini.
- Built an OpenRouter client for GPT, Claude, and Gemini with SQLite caching, JSONL logs, and tests.
- Added templated scenarios and game-parameter hashing so each experiment run is seeded and reproducible.
- Generated paired local vs. tourist-trap restaurant scenarios across 11 cities as the manipulation testbed.

JAX ML Algorithms From-scratch machine learning and transformer implementations in JAX.

 github.com/zoravur/jax-ml-algorithms

- Built a GPT-2-style character-level transformer from scratch in JAX (RoPE, causal multi-head attention, weight tying) with a masked training loop and a KV-cache decoding path that generates ~19x faster than the naive forward pass.

DeepCode A model-agnostic AI coding agent harness with an evidence-backed memory system.

 github.com/zoravur/deepcode

- Built staged agent validation from compileall and Ruff through fast tests, the full suite, and an LLM judge.
- Added sandboxed bash execution with per-call timeouts and environment isolation for agent-driven shell tools.
- Built DeepCode, a self-hosting Python coding agent for any OpenAI-compatible API (DeepSeek by default).
- Designed SQLite memory with immutable event and artifact logs, cited claims, review, and conflict resolution.
- Stamped each agent turn with harness and worktree commit provenance to make coding sessions reproducible.
- Added secret redaction and macOS Keychain credential storage to protect sensitive data used by the harness.
- Implemented 20+ agent tools spanning code edits, sandboxed shell execution, web search, and Python inspection.
- Developed ~14K lines of Python across 74 source modules, backed by 631 collected tests and ~13.1K lines of tests.

EDUCATION

Bachelor of Computer Science, Pure Mathematics Minor

University of Waterloo

Distinction; Cumulative GPA: 83.98%

Conferred Jun 2024

Coursework: CS 747 - Software Verification Using Proof Assistants; CS 480 - Introduction to Machine Learning; CS 360 - Introduction to the Theory of Computing; STAT 230 - Probability

SKILLS

Machine learning and data | JAX, PyTorch, transformers, reinforcement learning, mechanistic interpretability, NumPy, CNNs, Whisper

Languages | Python, C++, Lean 4, Haskell, Rust, TypeScript, SQL

Development practices | coding harnesses, agent orchestration, prompt engineering, TDD

Infrastructure and tooling | Linux, Docker, SQLite, PostgreSQL, Git, uv, AWS